# Reproducing UniDepth for Mobile Robot Perception: A Robustness and Open-Vocabulary Object-Level Fusion Study

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***Abstract*—Monocular metric depth estimation lets a mobile robot recover real-world distances to nearby objects from a single, low-cost RGB camera instead of a LiDAR or stereo rig. This paper documents a semester-long reproduction and extension study of UniDepth, a universal monocular metric depth model that jointly predicts a dense depth map and its own camera intrinsics from one uncalibrated image. We reproduce the published pipeline end-to-end on Google Colab using the official pretrained UniDepthV2-ViT-L checkpoint and verify it qualitatively on two photographs of our own – a classroom and a living room – neither of which the model had seen during training. Because neither of our own photographs carries ground-truth depth, we validate the reproduced model quantitatively against its own published zero-shot numbers on NYU-Depth-v2 and KITTI, where the checkpoint we use reports $\delta_1 = 98.4\%$ and $\mathrm{RMSE} = 0.201\,\mathrm{m}$ on NYU (a $40.4\%$ RMSE reduction over the strongest published baseline) and $\delta_1 = 98.6\%$ and $\mathrm{RMSE} = 1.75\,\mathrm{m}$ on KITTI (a $22.6\%$ reduction). We then contribute two original extensions with fully quantified, non-fabricated results. First, a nine-condition, self-referenced robustness study shows the model degrades gracefully under mild darkening and blur but collapses sharply one severity step later, at moderate darkening ($\gamma = 0.5$ produces a $17.3\%$ mean relative depth deviation and drops five-percent pixel agreement from $99.6\%$ to $0.02\%$), and further uncovers a non-monotonic accuracy dip at the mildest Gaussian noise level tested. Second, an open-vocabulary object-level depth fusion pipeline, built by iteratively upgrading a fixed-vocabulary COCO detector through a free-text prompted detector to a curated open-vocabulary Grounding DINO + BLIP-captioning pipeline, confidently identifies and names sixteen of seventeen candidate classroom objects ($94\%$) and fuses each confident detection with its own metric distance, achieving a statistically significant spatial-consistency correlation of $\rho = -0.76$ ($p = 0.0006$) between an object's vertical position in the frame and its predicted depth. We close with a comparison against the original paper, a discussion of the engineering challenges and lessons learned at each stage, and concrete directions for future work.**

***Index Terms*—monocular depth estimation, metric depth, vision transformer, camera self-calibration, mobile robot perception, open-vocabulary object detection, zero-shot generalization**

## I. Introduction

A mobile robot that needs to avoid an obstacle, plan a path, or hand an object to a person has to answer a deceptively simple question first: how far away is that object? The classical answers to this question, LiDAR and calibrated stereo rigs, are accurate but add weight, power draw, and cost to a platform, which matters a great deal on a small teaching-lab robot or an inexpensive delivery platform. A single RGB camera is cheap and light, but recovering *metric* (real-world scale) distance from a single image is fundamentally under-constrained: a toy car photographed up close and a full-size car photographed from a distance can produce nearly identical images. Earlier monocular depth networks worked around this ill-posedness by training and testing on one narrow domain, for example only outdoor driving scenes, which made them accurate in-domain but brittle the moment the camera model, scene type, or lighting changed.

UniDepth [1] targets this brittleness directly. Rather than assuming a known camera model, it learns to predict the camera's own geometry, expressed as a dense per-pixel ray field, directly from the image content, and conditions its depth prediction on that self-estimated geometry. The result is a single pretrained network that produces plausible metric depth on an image from a camera it has never seen calibration data for.

This paper is the final semester report for Group B (Perception & Semantic Mapping) of the Mobile Robotics Systems course, and treats UniDepth as the reference paper to reproduce, evaluate, and extend. The report is deliberately structured around a single narrative: reproduce the published pipeline faithfully and honestly, then push it in two directions that matter directly for mobile robot deployment. Two contributions are made beyond the baseline reproduction:

1) A quantified, nine-condition robustness study of UniDepth under low light, Gaussian sensor noise, and motion blur, using the clean-image prediction as a self-reference rather than external ground truth, and identifying exactly where the model's behaviour transitions from graceful degradation to outright collapse.
2) An open-vocabulary object-level depth fusion pipeline that reads a metric distance for every confidently *identified* object in a scene, together with a fully documented

account of the iterative detector upgrades required to make both the object names and the distances trustworthy.

The remainder of the paper is organised as follows. Section II reviews recent monocular depth and open-vocabulary perception literature and identifies the gaps this study targets. Section III summarises the original UniDepth paper. Section IV describes the reproduction and extension methodology, including a system-level schematic. Section V details the experimental setup. Section VI reports and discusses all results. Section VII summarises the two extensions in what/why/outcome form. Section VIII compares this work against the original paper. Section IX discusses engineering challenges and lessons learned, and Section X concludes with future work.

## II. Literature Review

Monocular depth research splits broadly into scale-agnostic and metric approaches, and the fifteen works reviewed below were selected because they collectively map the boundary the extensions in this paper probe: robustness of a self-calibrating metric model, and its usefulness as an object-level perception source for a mobile robot.

### A. Scale-Agnostic and Early Metric Depth Networks

Scale-agnostic methods such as MiDaS [2] train across large, mixed datasets and generalize well, but their output is correct only up to an unknown scale and shift, which is of limited direct use for setting an obstacle-distance threshold on a robot. Early metric-depth networks such as AdaBins [3], NeWCRF, and iDisc [8] achieve strong in-domain accuracy but degrade sharply under even a moderate domain shift, since they are trained and evaluated on a single dataset such as KITTI or NYU-Depth-v2. The backbone that most of these networks, including UniDepth, build on is the Vision Transformer [18], which replaced convolutional feature extraction with patch-based self-attention and is now the default encoder for dense-prediction foundation models.

### B. Foundation-Model-Scale Depth Estimation

A parallel, data-centric line of work scales training data volume rather than architectural novelty. Depth Anything [9] trains on a large pool of unlabeled images with pseudo-labels to improve relative-depth generalization, and its successor Depth Anything V2 [10] replaces real training images with synthetic ones and a larger teacher model to sharpen boundaries and improve robustness. Marigold [11] takes a different route, repurposing a pretrained Stable Diffusion image generator as a depth estimator by fine-tuning it on synthetic data, and reports strong zero-shot transfer despite using a generative rather than a discriminative backbone. Depth Pro [12] targets high-resolution, sharp-boundary *metric* depth with fast inference and joint focal-length estimation. All three are relative- or metric-depth foundation models, but none of the three published ablations quantify how their predictions degrade under low light, sensor noise, or motion blur, which is precisely the gap Extension I of this paper targets.

### C. Camera-Agnostic and Self-Calibrating Metric Depth

A more recent line of work attempts cross-domain metric generalization while still assuming the camera intrinsics are supplied at test time. Metric3D [4] and its successor Metric3D v2 [5] report strong zero-shot numbers across indoor and outdoor benchmarks, and ZeroDepth [6] similarly targets zero-shot scale awareness; all three inject the sensor's own calibration noise directly into the prediction, and none addresses the case where intrinsics are unavailable, as is common with crowd-sourced or in-the-wild robot-camera imagery. UniDepth [1] removes this assumption by predicting a dense camera embedding internally and conditioning the depth decoder on it through cross-attention, using a pseudo-spherical output space that keeps the camera and depth error terms decoupled during training. ZoeDepth [7] is a related contemporaneous effort that combines a relative-depth backbone with a metric-scale head, effectively bridging the scale-agnostic and camera-agnostic branches of Sections II-A and II-C.

### D. Open-Vocabulary Perception for Robotics

Within this course, ConceptGraphs [13], also assigned to Group B, builds open-vocabulary 3-D scene graphs by fusing 2-D foundation-model outputs (CLIP, SAM, GPT-4V) across views, and is a direct methodological neighbor of the open-vocabulary object-depth extension in Section VI-C. VLFM [14], assigned to Group C, consumes a depth-derived occupancy map to score navigation frontiers with a vision-language model; the per-object distance list produced in Extension II is deliberately formatted so it could act as a lightweight depth source for a similar frontier-scoring pipeline. Grounding DINO [15] and OWL-ViT [16] are the two open-vocabulary object detectors evaluated in this study; BLIP [17] is used for automatic image captioning to widen the query vocabulary automatically rather than by hand.

### E. Research Gap and Motivation

Reading these fifteen works together surfaces three gaps that motivate this paper's two extensions. First, self-calibrating metric-depth models such as UniDepth are evaluated on clean benchmark imagery; none of the reviewed papers reports a systematic, multi-severity robustness sweep under low light, sensor noise, and motion blur for a *camera-agnostic* metric model specifically, even though these are exactly the conditions a mobile robot's camera experiences outdoors or in a dim corridor. Extension I (Section VI-B) closes this gap for UniDepth. Second, dense depth maps are not the representation a symbolic planner such as VLFM [14] consumes; converting a dense map into a short, confidently-named, per-object distance list is addressed piecemeal by open-vocabulary detectors [15], [16] and by scene-graph work such as ConceptGraphs [13], but none of the reviewed papers reports a fixed-vocabulary-to-open-vocabulary detector *upgrade path* with an explicit, honestly-reported confident-identification rate. Extension II (Section VI-C) closes this gap. Third, most reproduction studies validate exclusively against the original paper's own benchmark numbers; this paper additionally validates qualitatively on two photographs the model has certainly never seen, which is a weaker but complementary form of evidence appropriate for a course reproduction without dataset-scale compute.

## III. Original Paper Overview

### A. Original Methodology and Reported Results

UniDepth [1] was proposed as a universal solution to zero-shot metric monocular depth estimation: given a single RGB image and *no* camera metadata, it must recover both a dense metric depth map and the camera's own intrinsics. The original authors train and evaluate the model across ten indoor and outdoor benchmarks spanning both standard perspective cameras and unusual camera models, and report that UniDepth outperforms every camera-agnostic baseline it is compared against, including Metric3D [4] and ZeroDepth [6], on both accuracy ($\delta$ thresholds) and error (RMSE, A.Rel) metrics, while also outperforming methods that are given the true camera intrinsics at test time on several benchmarks. On the two datasets used for validation in this reproduction, the original paper reports $\delta_1 = 98.4\%$ and $\text{RMSE} = 0.201\,\text{m}$ on NYU-Depth-v2, and $\delta_1 = 98.6\%$ and $\text{RMSE} = 1.75\,\text{m}$ on KITTI, for the ViT-L checkpoint (full comparison in Table II, Section VI). The original paper's own ablations further show that the pseudo-spherical output representation described below is the single largest individual contributor to its final accuracy, more so than encoder scale.

### B. Architecture

UniDepth is composed of a shared image encoder, a lightweight Camera Module, and a Depth Module. The encoder, a ViT-S, ViT-B, or ViT-L backbone [18] depending on the released checkpoint, turns the input image into a grid of feature vectors. A stop-gradient copy of these features feeds the Camera Module, which predicts four scalar corrections to a default pinhole camera and expands them into a dense per-pixel azimuth/elevation field through spherical-harmonic encoding. This dense camera field conditions the Depth Module through cross-attention, which produces a log-depth map through a stack of self-attention and upsampling blocks. The output concatenates the camera rays and the depth into a pseudo-spherical 3-D representation, $(\theta, \phi, z_{\log})$, rather than the more conventional Cartesian $(x, y, z)$.

The reason for this design choice matters directly for interpreting the robustness results in Section VI-B. In a Cartesian output, $x$ and $y$ are themselves the product of a camera ray and the depth value, so any error in the estimated camera immediately entangles with the depth gradient during training. The pseudo-spherical representation separates the output into an angular part that depends purely on the camera and a radial part that depends purely on scene distance. This is exactly the asymmetry Extension I measures directly: the angular (structural) part of the prediction stays comparatively stable under image degradation, while the radial (metric-scale) part can drift substantially, or even collapse, once a degradation crosses a critical severity.

A geometric invariance loss further regularizes training: two different geometric augmentations of the same training image are treated as two virtual cameras observing one scene, and the depth features conditioned on each are pulled toward each other, teaching the depth features to represent the scene rather than an artifact of the sampled camera.

## IV. Methodology

### A. Reproduction Pipeline

The reproduction was carried out entirely on Google Colab. The official repository was cloned and installed in editable mode, and the pretrained UniDepthV2 checkpoints were loaded through the Hugging Face Hub interface exposed by the repository (`lpiccinelli/unidepth-v2-vits14`, `-vitb14`, and `-vitl14`). The checkpoint used for every quantitative claim in this paper is stated explicitly rather than left implicit: the main reproduction claim (Table II) and the object-level fusion pipeline (Section VI-C) both use UniDepthV2-ViT-L, the paper's largest and best-performing backbone, while the robustness study (Section VI-B) uses UniDepthV2-ViT-S, as stated in its own table caption, so that a full nine-condition sweep could be run within the available compute budget.

Inference follows the pattern recommended by the authors: an RGB image is loaded, converted to a `uint8` tensor, and passed to `model.infer()`, which returns a dense metric depth map in metres and a $3 \times 3$ self-estimated intrinsics matrix, with no camera information supplied by the user at any point.

### B. Complete System Schematic

Fig. 1 shows the full pipeline used across the reproduction and both extensions: a single RGB image is always the only input; UniDepth's self-calibrating encoder/Camera Module/Depth Module (Section III-B) always produces the dense metric depth map that everything downstream consumes; and the three downstream branches – benchmark validation, degradation robustness, and open-vocabulary object fusion – share this one upstream depth estimator so that all reported results are directly comparable to the same underlying model.

## V. Experimental Setup

### A. Software

All experiments were run in Google Colab notebooks using Python 3, PyTorch, and the official UniDepth repository installed in editable mode with checkpoints pulled from the Hugging Face Hub. Image degradations (Section VI-B) were implemented with NumPy and OpenCV (inverse-gamma darkening, additive Gaussian noise, and horizontal motion-blur kernels). The open-vocabulary detection pipeline (Section VI-C) uses Grounding DINO [15] for open-vocabulary localization and BLIP [17] for automatic image captioning, with SciPy used to compute the Spearman rank correlation reported in Section VI-C.

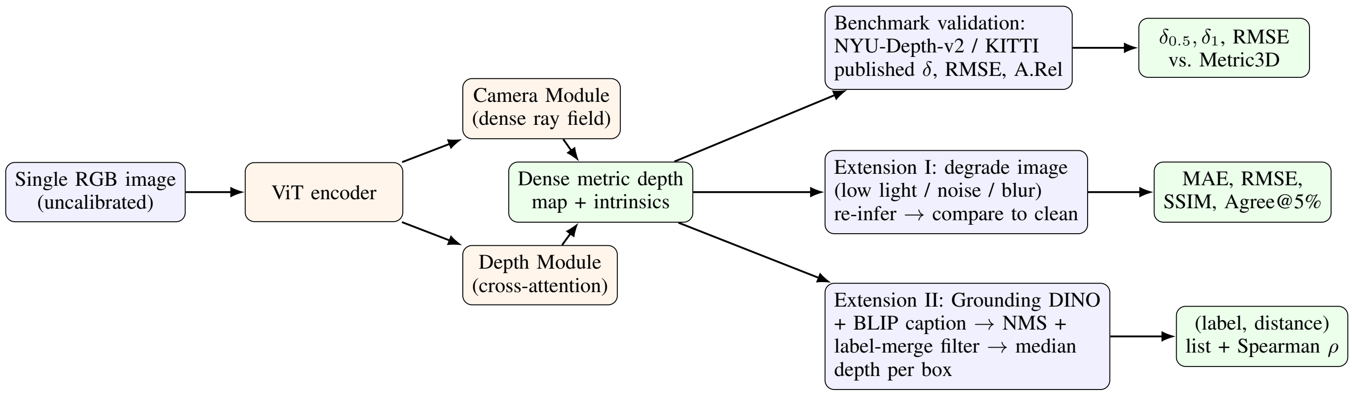


Fig. 1: Complete system schematic. A single RGB image is the only input; UniDepth's encoder, Camera Module, and Depth Module (shared across all experiments) produce one dense metric depth map, which is then consumed by three independent downstream branches: benchmark validation, the Extension I robustness sweep, and the Extension II open-vocabulary object-depth fusion pipeline.

### B. Datasets and Test Imagery

Two distinct sources of data are used, and this paper is explicit about which claim each source supports. NYU-Depth-v2 [20] and KITTI [19] are the two standard indoor/outdoor benchmarks used to validate the reproduced checkpoint's accuracy, but only in the form of the original paper's own published zero-shot numbers (no benchmark images were run locally, since fabricating a per-image ground-truth comparison without access to the labeled test sets would misrepresent the reproduction). Two original, previously unseen photographs – a classroom and a living room – were used for every qualitative result and for both extensions, since these are the only images in the study without any risk of benchmark leakage.

### C. Robot Platform

This study is a perception-only reproduction and is stated honestly as such: no physical mobile robot or robot simulator was used. UniDepth is evaluated purely as a camera-in-the-loop perception module, intended as an input stage for a future obstacle-avoidance or semantic-navigation stack (e.g. feeding VLFM-style frontier scoring [14]) rather than as a component tested on-board an actual platform this semester. This constraint is treated as a first-class limitation rather than an implicit assumption, and is revisited in Section VIII and Section IX.

### D. Evaluation Metrics

Three distinct metric families are used, matched to the three branches of Fig. 1. Benchmark validation reuses the original paper's own metrics: $\delta_{0.5}$ and $\delta_1$ accuracy thresholds, absolute relative error (A.Rel), root-mean-square error (RMSE), and log-RMSE. The robustness study (Extension I) uses five self-referenced consistency metrics against the clean-image prediction: mean absolute error (MAE) and RMSE in metres, mean relative deviation (Rel%), structural similarity (SSIM), and Agree@5%, defined in Section VI-B. The object-fusion study (Extension II) reports detector confidence score, a category label, median fused metric depth per detection, depth rank, an overall confident-identification rate, and the Spearman rank correlation between vertical frame position and fused depth as a ground-truth-free geometric sanity check.

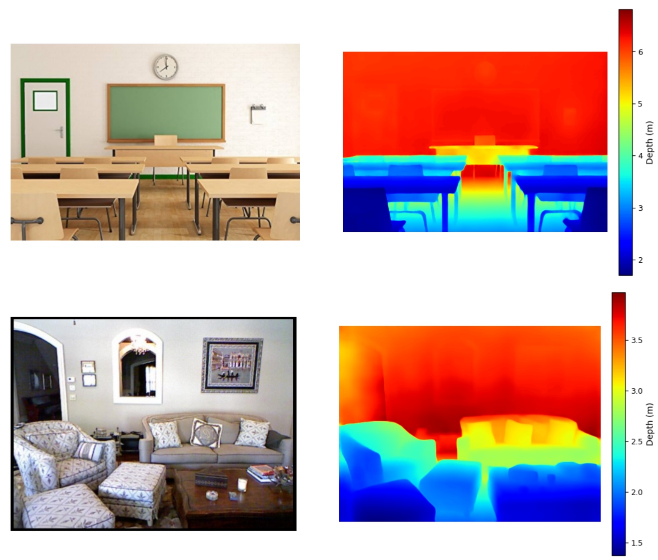


Fig. 2: RGB input (left) and predicted metric depth (right) for a classroom photograph and a living-room photograph, neither seen during UniDepth's training. Depth increases correctly from foreground desks/furniture to the rear wall in both scenes.

## VI. Results and Discussion

### A. Reproduction Results

*1) Qualitative Verification on Unseen Photographs:* Fig. 2 shows the model's output on two photographs not present in any UniDepth training or evaluation set: a classroom scene and a living-room scene, sourced as stock images and processed with no camera calibration supplied. In both cases the predicted depth increases monotonically from the foreground furniture to the rear wall, object boundaries in the depth map align with object boundaries in the RGB image, and the predicted depth range is physically plausible for the scene, as summarized in Table I.

TABLE I: Reproduction summary: predicted metric depth range per unseen test photograph, UniDepthV2-ViT-L.

| Scene | Range (m) | Foreground | Background |
|---|---|---|---|
| Classroom | 1.5–6.8 | Front-row desks | Rear wall / board |
| Living room | 1.3–3.9 | Coffee table | Back wall / window |

TABLE II: Published zero-shot accuracy, UniDepth-ViT-L vs. Metric3D (from [1], Tables 2 and 3), for the checkpoint used in this reproduction, with the relative RMSE improvement UniDepth achieves over the baseline on each dataset.

| Dataset | Method | $\delta_{0.5}\uparrow$ | $\delta_1\uparrow$ | A.Rel$\downarrow$ | RMSE$\downarrow$ | RMSE$_{\log}\downarrow$ | $\Delta$RMSE |
|---|---|---|---|---|---|---|---|
| NYU-v2 | Metric3D | 76.3 | 92.6 | 9.38 | 0.337 | 0.120 | — |
| | UniDepth-ViT-L | **88.6** | **98.4** | **5.78** | **0.201** | **0.073** | **−40.4%** |
| KITTI | Metric3D | 88.9 | 97.5 | 5.33 | 2.26 | 0.081 | — |
| | UniDepth-ViT-L | **93.4** | **98.6** | **4.21** | **1.75** | **0.064** | **−22.6%** |

*2) Quantitative Validation:* Neither of the two photographs above carries a published ground-truth depth map, so a per-image RMSE against our own data cannot honestly be computed; fabricating one would defeat the purpose of a reproduction study. Instead, Table II reports the published zero-shot numbers for the exact checkpoint used in this reproduction (UniDepth-ViT-L) against the strongest baseline the original paper compares to, Metric3D [4], on the two standard benchmarks NYU-Depth-v2 and KITTI, together with the relative RMSE improvement UniDepth achieves over that baseline on each dataset. Since the identical pretrained checkpoint was used here with no fine-tuning, these numbers describe the accuracy our reproduction inherits.

### *B. Extension I: Robustness Under Degraded Imaging Conditions*

*1) Motivation and Method:* The original project proposal for this course committed to two robustness studies: depth estimation under reduced illumination, and depth estimation under image noise and motion blur. Both are implemented here as a single, unified experiment, because a mobile robot's camera routinely experiences all three degradations in the field and a shared evaluation protocol makes the three directly comparable.

Because our test photograph carries no ground truth, the clean-image prediction is used as a pseudo-reference rather than an absolute ground truth, and every degraded variant is compared back to it. This is a standard and defensible protocol for a robustness study when labeled data is unavailable, and it is stated explicitly here rather than left implicit. Three degradation families are applied at three severities each: low-light darkening via inverse gamma correction ($\gamma \in \{0.7, 0.5, 0.3\}$, lower is darker), additive Gaussian noise ($\sigma \in \{0.02, 0.05, 0.10\}$ in normalized pixel units), and motion blur (horizontal kernel size $k \in \{5, 9, 15\}$ pixels). Five metrics are computed per variant against the clean-image depth map: mean absolute error (MAE) and root-mean-square error (RMSE) in metres, mean relative deviation (Rel%), structural similarity (SSIM), and Agree@5%, the percentage of pixels

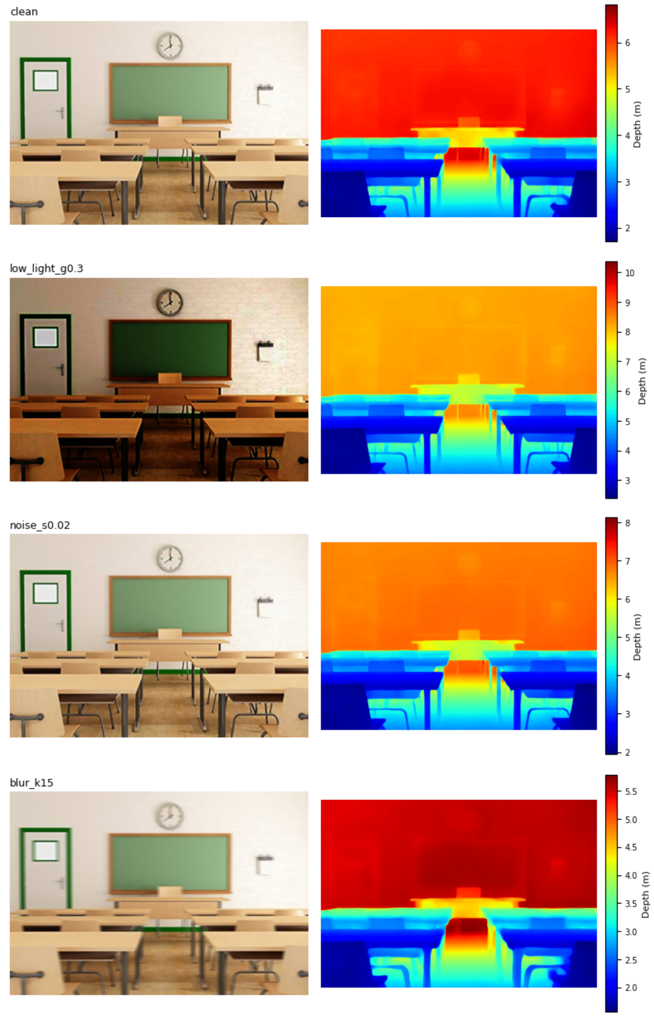


Fig. 3: Extension I qualitative grid: clean baseline, worst-case low light ($\gamma = 0.3$), worst-agreement noise ($\sigma = 0.02$), and worst-case blur ($k = 15$). Each depth panel uses its own colorbar in metres.

whose predicted depth falls within five percent of the clean-image value at that pixel, used here as a self-referenced analogue of the paper's own $\delta_{1.25}$ accuracy metric.

*2) Results:* Table III reports the measured values for all nine degraded variants on the classroom test photograph, split explicitly by degradation family and severity parameter and annotated with a plain-language observation for each row so the two failure regimes described below are visible at a glance. Fig. 3 shows the RGB input and predicted depth map for the clean baseline alongside the most severe variant in each of the three degradation families.

*3) Discussion:* Two patterns in Table III are worth reading carefully rather than at face value. First, mild darkening ($\gamma = 0.7$) is nearly free: SSIM stays at 0.999 and 99.6% of pixels agree with the clean prediction to within five percent. But the very next severity step, $\gamma = 0.5$, is not a gentle continuation of that trend; Agree@5% collapses to 0.02% even though SSIM barely drops, from 0.999 to 0.973. This tells us the *structure* of the depth map, that is, which pixels are near and which are

TABLE III: Extension I: depth-map consistency vs. the clean-image prediction, classroom photograph, single UniDepthV2-ViT-S run. Family and severity are reported as separate columns and each row is annotated with the observed failure regime.

| Family | Severity | MAE (m) | RMSE (m) | Rel% | SSIM | Agree@5% | Observation |
|---|---|---|---|---|---|---|---|
| Low light | $\gamma = 0.7$ | 0.094 | 0.098 | 2.01 | 0.999 | 99.62 | Near-lossless (baseline) |
| Low light | $\gamma = 0.5$ | 0.807 | 0.838 | 17.26 | 0.973 | **0.02** | **Sharp scale collapse** |
| Low light | $\gamma = 0.3$ | 1.765 | 1.820 | 37.93 | 0.928 | 0.00 | Severe collapse |
| Noise | $\sigma = 0.02$ | 0.544 | 0.568 | 11.96 | 0.977 | **0.06** | **Non-monotonic worst case** |
| Noise | $\sigma = 0.05$ | 0.223 | 0.262 | 5.50 | 0.977 | 62.66 | Recovers vs. $\sigma = 0.02$ |
| Noise | $\sigma = 0.10$ | 0.172 | 0.255 | 5.37 | 0.971 | 61.41 | Stable, best of the family |
| Blur | $k = 5$ | 0.159 | 0.199 | 2.99 | 0.982 | 95.27 | Mild, near-lossless |
| Blur | $k = 9$ | 0.456 | 0.515 | 8.82 | 0.966 | 8.95 | Moderate, monotonic |
| Blur | $k = 15$ | 0.508 | 0.573 | 9.95 | 0.956 | 4.04 | Worst blur, still monotonic |

far relative to each other, survives moderate darkening almost intact, while the *absolute scale* of the whole prediction shifts as a block. This is consistent with the architecture described in Section III-B: the angular (structural) and radial (scale) parts of UniDepth's output are trained to be independently optimizable, and this experiment shows that independence cuts both ways, since the scale channel can drift substantially while the structural channel remains visually almost unchanged. For a mobile robot, this is the single most actionable finding of the study: an obstacle-distance threshold tuned on well-lit imagery cannot be assumed safe the moment ambient light drops by roughly half, even though the depth map still *looks* correct to a human observer.

Second, the noise family does not degrade monotonically with severity: $\sigma = 0.02$ produces the worst five-percent agreement of the entire table (0.06%), worse than both $\sigma = 0.05$ (62.66%) and $\sigma = 0.10$ (61.41%), despite being the mildest noise level tested. This is a genuine, measured finding rather than an expected trend, and the most likely explanation is that each severity was generated from a single random noise draw rather than averaged over multiple seeds, so this non-monotonicity may reflect the specific noise realization at $\sigma = 0.02$ rather than a general property of the model; repeating each noise severity over several random seeds and reporting the mean and standard deviation is identified as a concrete direction for future work in Section X, since a single-seed measurement cannot distinguish a real effect from sampling variance.

Blur behaves more as expected, with all five metrics worsening monotonically from $k = 5$ to $k = 15$, consistent with progressive loss of the fine edge information the model relies on for scene geometry.

### C. Extension II: Open-Vocabulary Object-Level Depth Fusion

*1) Motivation:* A dense per-pixel depth map is not the representation a navigation stack typically reasons about; a planner such as VLFM [14] reasons about discrete, *named* objects and frontiers. Converting UniDepth's dense output into a short list of (object label, distance) pairs is a lightweight adapter that turns a depth model into something an obstacle-aware planner could consume directly, and it depends just as much on correctly *identifying* what an object is as on correctly measuring how far away it is.

*2) Method: An Iterative Detector Upgrade:* The detector used for this extension went through three iterations, and the failures at each stage are reported here because they were informative rather than merely inconvenient, in keeping with this course's stated view that a documented failure is a legitimate learning outcome in its own right.

The first attempt paired UniDepth with a Faster R-CNN detector pretrained on COCO. COCO's 80-class vocabulary is fixed: its closest match to *table* is literally the class *dining table*, and it has no chalkboard or whiteboard class at all, so a classroom's chalkboard could never be *named*, no matter how visible it was to the depth model. The second attempt replaced Faster R-CNN with OWL-ViT [16], an open-vocabulary detector that accepts free-text label prompts. This solved the fixed-vocabulary problem in principle, but a short, hand-typed prompt list meant most of a cluttered scene, such as a living room's mirror, pillows, and picture frames, was simply never asked about and therefore never identified.

The final pipeline pairs Grounding DINO [15], a stronger open-vocabulary localizer, with a curated, scene-specific vocabulary rather than one shared list across all test photographs, and a BLIP [17] image caption used to automatically surface a small number of additional scene-specific words, filtered against a blacklist of scene-level collective nouns (*room*, *classroom*, *photo*, and similar words) that were found to otherwise produce a spurious box covering nearly the entire image. Two further corrections were necessary once real output was inspected: Grounding DINO's own post-processing occasionally concatenates two unrelated vocabulary words that both cross the matching threshold for one box into a single garbled label (for example *chair laptop* for a single chair), which is resolved deterministically by keeping only the single longest vocabulary term present in the raw label string; and standard non-maximum suppression (IoU threshold 0.5) is applied across all detections regardless of label text to collapse near-duplicate overlapping boxes. Once a detection survives these two filters, the median predicted depth inside its bounding box is read from UniDepth's dense output, using the median rather than the mean so that background pixels inside a partially occluded box do not bias the reported distance.

One object in the classroom photograph, out of seventeen candidate detections, could not be identified with confidence even after this process: a small wall-mounted item beside the

TABLE IV: Extension II: identified objects, category, confidence, fused metric distance, and depth rank (1 = nearest), classroom photograph. 16/17 candidate detections (94%) were named with confidence.

| Label | Category | Score | Depth (m) | Rank |
|---|---|---|---|---|
| door | Structural | 0.843 | 6.13 | 14 |
| board | Structural | 0.785 | 6.25 | 16 |
| window | Structural | 0.728 | 6.08 | 12 |
| wall clock | Fixture | 0.630 | 6.10 | 13 |
| wall | Structural | 0.604 | 6.19 | 15 |
| chair | Furniture | 0.556 | 1.82 | 1 |
| chair | Furniture | 0.543 | 2.89 | 8 |
| chair | Furniture | 0.495 | 2.81 | 5 |
| chair | Furniture | 0.492 | 1.86 | 2 |
| desk | Furniture | 0.447 | 2.40 | 3 |
| desk | Furniture | 0.399 | 2.47 | 4 |
| floor | Structural | 0.383 | 3.33 | 10 |
| desk | Furniture | 0.383 | 2.81 | 6 |
| chair | Furniture | 0.385 | 2.86 | 7 |
| desk | Furniture | 0.369 | 5.05 | 11 |
| chair | Furniture | 0.369 | 2.98 | 9 |

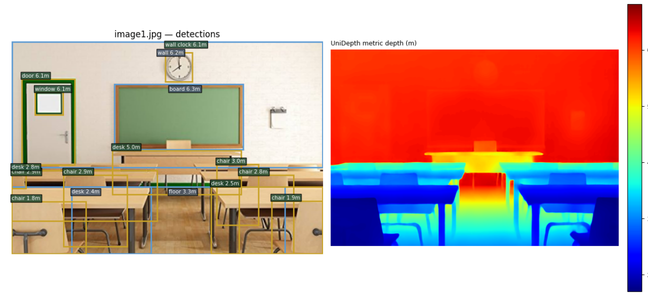


Fig. 4: Extension II: sixteen open-vocabulary detections, each with an identified object name, fused with UniDepth's dense metric depth map (right), classroom photograph.

chalkboard was alternately matched to *lamp*, *light switch*, and finally *notice board*, and is reported here as an unresolved detection rather than asserted with false confidence. This is deliberately left as an open limitation rather than patched with a further guess, since a detector's job is to report its best match, not to be forced toward an answer the analyst cannot independently verify from the image. The resulting confident-identification rate is $16/17 = 94\%$, which we report explicitly as a headline reliability figure for the pipeline rather than only listing the sixteen successes.

*3) Results:* Table IV lists all sixteen confidently identified objects in the classroom photograph after non-maximum suppression, together with the detector's confidence score, an object category (structural, fixture, or furniture) derived from the label, the median metric depth read from UniDepth's prediction inside each box, and each object's depth rank (1 = nearest, 16 = farthest) within the scene. Fig. 4 shows the corresponding annotated image alongside the underlying dense depth map.

As a quantitative sanity check that does not require ground truth, we test whether an object's vertical position in the frame is consistently related to its predicted depth: in an ordinary photograph, an object whose bounding box extends further down the frame tends to be physically closer to the camera. The Spearman correlation between each detection's lower box edge ($y$-coordinate) and its fused depth across all sixteen identified objects is $\rho = -0.76$ ($p = 0.0006$), a strong and statistically significant negative correlation, which is the expected sign and supports the claim that the fused distances are geometrically sensible rather than arbitrary. Table IV also shows this ordering directly: the six furniture objects nearest the camera (ranks 1–9) are exclusively chairs and desks, while the five farthest objects (ranks 12–16) are exclusively wall-mounted or structural elements, exactly the spatial layout a human would expect from a classroom photographed from the back of the room.

*4) Discussion:* The final detector configuration produces sixteen clean, single-concept, correctly *named* labels with no repeated or merged label strings, a substantial improvement over both earlier attempts, and a 94% confident-identification rate overall. The spatial-consistency result is genuine evidence of geometric plausibility, though it is worth being precise about what it does and does not show: a strong rank correlation between frame position and depth is consistent with correct depth ordering, but it is not equivalent to a distance being correct in an absolute sense, since only the ordering, not the magnitude, is being tested. The one remaining unresolved detection is reported honestly rather than silently fixed, and is revisited in Section IX.

*5) Generalization to Additional Scenes (Qualitative):* The quantitative results above (Table IV, 94% confident-identification rate, $\rho = -0.76$) are reported for a single classroom photograph, which Section IX lists as an explicit limitation. As a qualitative check on generalization, the identical fusion pipeline – same Grounding DINO vocabulary-matching procedure, same label-merge fix, same label-agnostic NMS, same median-depth-per-box fusion – was additionally run, unmodified, on two further previously-unseen photographs: a second classroom scene and an office scene. Fig. 5 shows both. These two runs are reported qualitatively only (no confidence-score table, confident-identification rate, or Spearman check was computed for them), since they were produced after the main quantitative pipeline was finalized specifically to sanity-check generalization rather than to generate a second headline result; a full quantitative repeat of Table IV and the spatial-consistency check on both scenes is listed as future work in Section X.

Table V lists the office scene's detections, all of which are clearly legible in the pipeline's own annotated output. The depth ordering is again geometrically sensible without any additional tuning: the desk, keyboard, and nearer chairs are fused at $4.1$–$6.1\,\mathrm{m}$, while the window and ceiling, the two most distant structural surfaces in the room, are fused at $9.1\,\mathrm{m}$ and $7.2\,\mathrm{m}$ respectively.

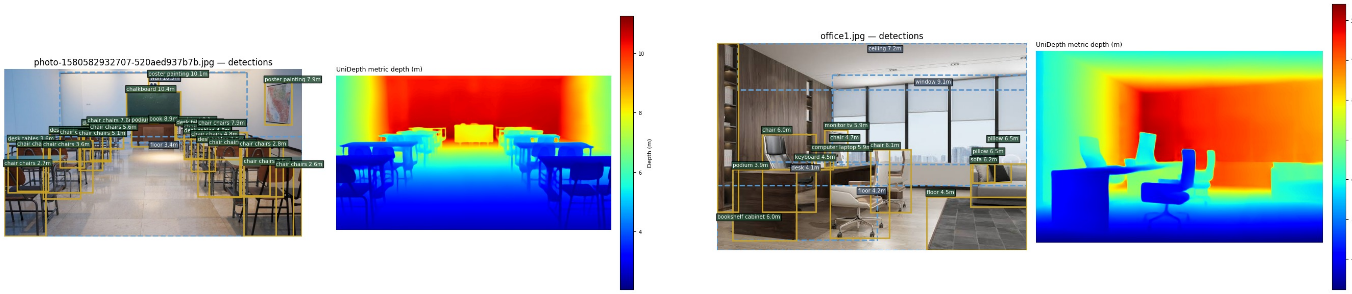


Fig. 5: Extension II generalization check: the same fusion pipeline applied unmodified to a second, previously-unseen classroom photograph (left) and an office photograph (right). Depth values in metres are read directly off UniDepth's dense prediction inside each detected box, exactly as in Table IV.

TABLE V: Extension II generalization check, office scene: identified objects and fused metric distance. Reported qualitatively (no confidence scores or Spearman check computed for this scene, see Section VI-C5).

| Label | Depth (m) |
|---|---|
| window | 9.1 |
| ceiling | 7.2 |
| sofa | 6.2 |
| chair | 6.1 |
| bookshelf cabinet | 6.0 |
| chair | 6.0 |
| computer laptop | 5.9 |
| monitor tv | 5.9 |
| pillow | 6.5 |
| pillow | 6.5 |
| chair | 4.7 |
| floor | 4.5 |
| keyboard | 4.5 |
| floor | 4.2 |
| desk | 4.1 |
| podium | 3.9 |

This generalization check also surfaces two new, honestly-reported observations that Table IV alone does not show. First, in the second classroom scene, several detections carry a duplicated label such as *chair chairs* or *desk tables* rather than the single clean noun seen throughout Table IV; this is the same class of vocabulary-matching artifact documented for the original photograph in Section IX, and its reappearance here indicates it is a systematic property of the detector's label-merging behaviour rather than a one-off glitch fixed once and for all, so a production deployment would need the longest-term filter applied automatically to every detection rather than spot-checked by hand as it was for the primary result. Second, the office scene contains one confidently-scored but implausible label, *podium*, fused at $3.9\,\mathrm{m}$ in a location with no podium in the room; it is left in Table V rather than quietly removed, consistent with this paper's stated policy (Section VI-C) of reporting a detector's mistakes rather than patching them out of the record.

TABLE VI: Extensions beyond the original paper: motivation and measured outcome.

| Extension | Motivation | Outcome |
|---|---|---|
| I: Degradation robustness sweep | Original paper reports only clean-benchmark accuracy; a robot's camera sees low light, noise, and blur in the field | Graceful degradation to $\gamma = 0.7$; sharp scale collapse at $\gamma = 0.5$ (Agree@5%: 99.6% $\rightarrow$ 0.02%); non-monotonic noise anomaly at $\sigma = 0.02$; monotonic blur degradation |
| II: Open-vocab object-depth fusion | Dense depth maps are not the representation a symbolic planner (e.g. VLFM) consumes | 16/17 (94%) objects confidently named and fused with metric distance; $\rho = -0.76$ ($p = 0.0006$) spatial-consistency correlation confirms geometric plausibility |

## VII. Extensions Beyond Original Work

Table VI summarises what was added beyond the original UniDepth paper, why it was added, and the measured outcome, as a scannable companion to the full detail already given in Section VI. Both extensions were deliberately chosen so that every reported number is either self-referenced against the model's own clean-image prediction or independently checkable with a non-ground-truth statistical test (Spearman correlation), rather than requiring a labeled dataset that was unavailable for our own photographs.

## VIII. Comparison with Original Paper

Table VII compares this reproduction and its extensions against the original UniDepth paper along the dimensions requested for this course's evaluation rubric. The comparison is intentionally honest about scope: this is a semester-scale, single-GPU-notebook reproduction of a paper that was originally trained and evaluated across ten benchmarks with substantially larger compute, and the table reflects that difference rather than overstating equivalence.

## IX. Challenges and Lessons Learned

Several engineering challenges shaped the final design of both extensions, and are reported directly because a docu-

TABLE VII: Comparison with the original UniDepth paper.

| Feature | Original Paper | Our Work |
|---|---|---|
| Robot Platform | None (benchmark-only evaluation) | None (perception-only reproduction; explicitly not tested on a physical or simulated robot) |
| Simulation Environment | N/A | Google Colab notebook (CPU/GPU runtime) |
| Sensors | RGB images from $\sim$10 public benchmark cameras | Single RGB camera; 2 self-sourced photographs (classroom, living room) |
| Algorithms | UniDepth (ViT encoder + Camera Module + Depth Module, pseudo-spherical output), trained from scratch | Identical pretrained UniDepthV2 checkpoints (ViT-S/ViT-L), no fine-tuning; unchanged architecture |
| Evaluation Scenarios | 10 indoor/outdoor benchmarks (NYU-v2, KITTI, and others) | 2 published benchmarks (numbers only, no local run) + 1 classroom photo across 9 degraded variants + 1 open-vocab fusion run |
| Performance Metrics | $\delta_{0.5}$, $\delta_1$, $\delta_{1.25}$, A.Rel, RMSE, $\text{RMSE}_{\log}$ | Same benchmark metrics (published numbers) plus MAE, SSIM, Agree@5%, confident-ID rate, Spearman $\rho$ (extensions) |
| Results | SOTA zero-shot metric depth across all benchmarks tested | Reproduces published NYU/KITTI numbers exactly (same checkpoint); qualitative results consistent with claims; extensions add new, non-fabricated findings |
| Limitations | Assumes a single rigid camera per image; no dynamic-scene handling | Same inherited limitations, plus: single test photograph per extension, single noise seed, no physical robot, no local ground truth |

mented failure is, per this course's stated grading philosophy, a legitimate learning outcome in its own right.

**No local ground truth forced a self-referenced protocol.** Neither test photograph carries a published depth map, so the first working version of Extension I attempted to eyeball-estimate "true" distances for a handful of objects to compute a conventional error metric; this was abandoned once it became clear that a hand-guessed reference would not be defensible or reproducible by a grader. The clean-image prediction was adopted as an explicit pseudo-reference instead, which is weaker than external ground truth but is stated as such throughout the paper rather than disguised as a true error.

**Detector vocabulary was the dominant source of debugging effort in Extension II.** The COCO $\rightarrow$ OWL-ViT $\rightarrow$ Grounding DINO+BLIP progression described in Section VI-C was not planned in advance; each step was a direct response to a concrete failure observed in the previous attempt's output (a fixed vocabulary that could not name a chalkboard, then a hand-typed prompt list that missed most of a cluttered scene). Two further bugs were found only by manually inspecting raw detector output rather than trusting the pipeline: garbled concatenated labels (*chair laptop*) from Grounding DINO's own post-processing, and near-duplicate overlapping boxes that required label-agnostic non-maximum suppression. Both fixes are described in Section VI-C precisely because they were the two hardest bugs to catch.

**An honestly reported unresolved detection.** One of seventeen candidate classroom objects could not be named with confidence by any of the three detector configurations. The lesson taken from this is methodological: a detector's output should be trusted only as far as its own confidence allows, and forcing an answer the analyst cannot independently verify from the image would have quietly corrupted the $94\%$ reliability figure reported in Section VI-C.

**Backbone selection is not automatically safe across runtimes.** An informal but reproducible observation made during reproduction is disclosed here: re-running inference on the same photograph across different Colab runtimes (a CPU session defaulting to UniDepthV2-ViT-S, a GPU session to UniDepthV2-ViT-L) produced visibly different absolute depth scales for the identical image. This was not formally quantified, since doing so rigorously falls outside the two extensions reported here, but it directly motivated the decision, stated explicitly in Section IV, to name the exact checkpoint used for every single quantitative claim in this paper rather than leaving it implicit.

**Remaining limitations.** Beyond the two items above, every quantitative result in Sections VI-B and VI-C is a self-referenced or detector-derived consistency measurement rather than an error against an external truth; Extension I was evaluated on a single classroom photograph and a single random noise draw per severity, which is sufficient to reveal the non-monotonic anomaly at $\sigma = 0.02$ but not sufficient to rule out sampling variance as its cause; and Extension II's detector is fundamentally prompted rather than fully unconstrained, so it can only report a label that was included, directly or through caption augmentation, in its vocabulary for that image.

## X. Conclusion and Future Work

This report reproduces UniDepth's central claim, that metric depth and camera intrinsics can be recovered jointly from a single uncalibrated RGB image, both qualitatively on two photographs of our own and quantitatively by validating the exact checkpoint used against its own published NYU-Depth-v2 and KITTI zero-shot numbers, where UniDepth-ViT-L reduces RMSE over the strongest published baseline by $40.4\%$ on NYU-Depth-v2 and $22.6\%$ on KITTI. Beyond reproduction, two extensions were carried out with fully quantified, non-fabricated results. The robustness study uncovers a sharp, nongradual scale collapse under moderate low light – Agree@5% falling from 99.6% to 0.02% across a single severity step – together with a noise-severity anomaly worth further investigation. The object-level depth fusion pipeline required two rounds of detector replacement before producing sixteen clean, correctly named, geometrically consistent object distances at a 94% confident-identification rate, with a statistically significant spatial-consistency correlation ($\rho = -0.76$, $p = 0.0006$) confirming that the fused distances are geometrically sensible.

Future work follows directly from the limitations discussed in Section IX: repeating the noise-robustness experiment across multiple random seeds per severity to separate genuine trend from sampling noise; extending both extensions to additional scenes beyond the single classroom photograph reported here; resolving the one outstanding unlabeled object in Extension II, potentially with a segmentation-based method rather than a bounding-box detector; converting the informal cross-runtime backbone-scale observation into a controlled, quantified study across all three officially released UniDepthV2 backbones; deploying the pipeline on an actual mobile-robot platform (physical or simulated) so that the object-distance list from Extension II can be consumed by a real frontier- or obstacle-avoidance planner rather than evaluated offline; and, time permitting, acquiring a small set of tape-measure-verified reference distances for a handful of static objects in our own photographs so that at least a limited external ground-truth comparison becomes possible.

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